

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY BHOPAL
DEPARTMENT OF ELECTRICAL ENGINEERING
MID TERM EXAMINATION-MARCH 2025

Course: B.Tech

Semester: IV

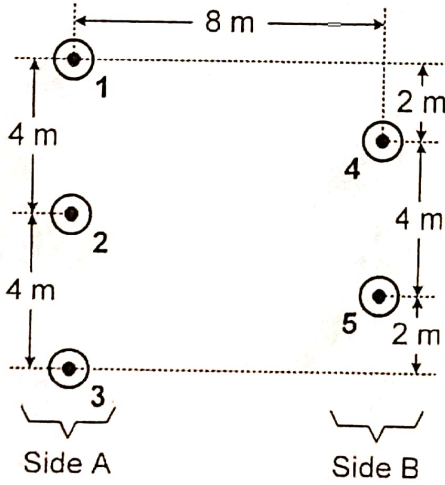
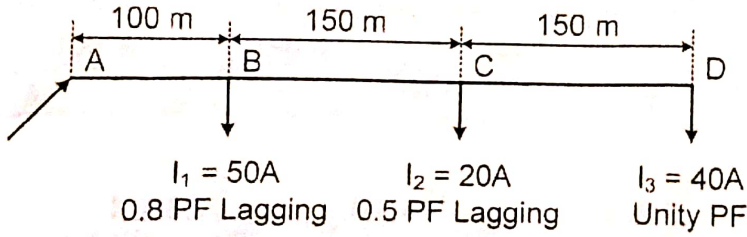
Subject Name: POWER SYSTEM

Subject Code: EE-221

Time Duration: 1 Hour 30 Min

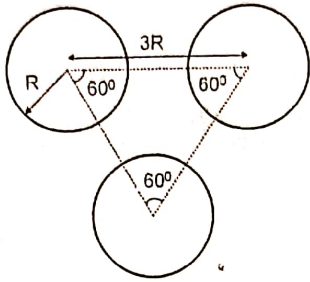
Maximum Marks: 20

Note: Assume necessary data if required.

S.No	Question	Marks	CO
1	The arrangement of conductor of a single-phase transmission line is shown in Fig. 1, wherein the forward circuit is composed of three solid wires 2.5 mm in radius and the return circuit of two-wires of radius 5 mm placed symmetrically with respect to the forward circuit. Find the inductance of each side of the line and that of the complete line.  Fig.1	4	CO-1
2	The loading on a distributor is shown in Fig.2. The distributor is a two-core cable for which the resistance and reactance are 0.25Ω and 0.125Ω per 1000 meters of cable run respectively. What should the voltage at point A to maintain 400 V at point D?  Fig.2	4	CO-4
3	A 2-wire DC ring distributor is 300 m long and is fed at 240 V at point A. At point B, 150 m from A, a load of 120 A is taken at C, 100 m in opposite direction, a load of 80 A is taken. If resistance per 100 m of single conductor is 0.03Ω, find - Current in each section of the distributor - Voltage at points B and C.	3	CO-4

Name of the Student _____

Scholar Number _____

4	A) Give a typical layout of a power system between generation and use of electrical power. B) A composite conductor consists of three conductors of radius R each. The conductors are arranged as shown in Fig.3. The geometric mean radius (GMR) in cm of the composite conductor is kR. The value of k is _____.  Fig.3	3 2	CO-1 CO-1
5	A 3-phase, 50 Hz transmission line has flat horizontal configuration with 3.5 m between adjacent conductors. The conductors are no. 2/0 hard-drawn seven strand copper (outside conductor diameter = 1.05 cm). The voltage of the line is 110 kV. Find the capacitance to neutral and charging current per km.	4	CO-1